

Anyonic Tensor Networks

Symmetries, Categories and Circuits

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CATEGORICAL NETWORKS

In the existing literature it is well known that all of the concepts we have seen up to now are modeled by a formalism called *tensor categories*:

1. fdVect_k and fdHilb tensors are the most trivial and motivating case and trivial build tensor categories.
2. $\text{Rep}(G)$ (for G finite) as in ?? are one of the primary motivations for construction fusion tensor categories. [ENO17].
3. All anyonic topological orders (i.e species of quasiparticles as in ??) can be classified by *modular tensor categories (MTCs)* [Kit06][RSW09].

Literature 1.0.1 (Tensor Categories). Grothendieck investigated what we now know as tensor categories as part of a larger study on *tannakian categories* to related the methods of representation theory (c.f. 3) to less related mathematical problems. Monoidal categories in turn were introduced by Benabou [empty citation] and MacLane [empty citation] under the name *bicategory* and *category with multiplication*, although an elementary construction of such categories is given by Tannaka [empty citation]

The name *tensor category* is due to Deligne [Del90] in "Catégories Tannakiennes" and its followup [Del02] BlaBlub

The correspondence between the penrose notation (i.e. planar diagrams) and the calculus provided by tensor categories is explained in great detail in Joyal and Street [JS91], although earlier accounts such as [RT91] already make use of such a string calculus.

We have also seen a method for evaluating networks built in $\text{Rep}(G)$ in ?? by decomposing a network into its fusion tree and its degeneracy spaces. As an immediate consequence, this yields an algorithm for evaluating generalized *anyonic* and *categorical* tensor networks, as in [Unf24, § 4][DH25] [Pfe12; Pfe+10] [Aye17] [Sch+20, § ?].

In light of this we provide a bottom-up construction of the theory of fusion categories in section 1.1, each concept will carry along an example for representations, a physical interpretation in terms of anyonic world-lines as well as the correspondence to the ATS systems of Penrose [Pen71]. We formalize the aforementioned decomposition in section 1.2 and list the algorithms necessary for achieving a tensor network out of these theories.

1.1 TENSOR & FUSION CATEGORIES

The general idea of a category is that it admits a sort of *interface*. As noticed, there are many similarities in data and structure between the entities we already dealt with, i.e. the tensor product of vector space, the tensor product of representations, of Hilbert spaces etc.... We can view all these concepts as a category with additional constraints added.

Definition 1.1.1. A *category* C consists of

- a set of *objects* $\text{ob}(C)$,
- for every pair $A, B \in \text{ob}(C)$, a *hom-set* $\text{Hom}_C(A, B)$ of *morphisms* $f : A \rightarrow B$,

there is one here from 1982 find what this does.

or tqfts?

Mention consisting of data?

- for every $A \in \text{ob}(C)$, an *identity morphism* $\text{id}_A : A \rightarrow A$,
- for every triple $A, B, C \in \text{ob}(C)$, a *composition map*

$$\circ : \text{Hom}_C(B, C) \times \text{Hom}_C(A, B) \longrightarrow \text{Hom}_C(A, C),$$

such that composition is *associative*, i.e. $(f \circ g) \circ h = f \circ (g \circ h)$ whenever defined, and *unital*, i.e. $f \circ \text{id}_A = f = \text{id}_B \circ f$ for all $f : A \rightarrow B$.

Example 1.1.2. Some examples here:

1. The category fdVect_k is the category of finite dimensional vector spaces and its morphisms are the linear operators between vector spaces.
2. The category fdHilb is the category whose objects are Hilbert spaces and whose morphisms are the bounded linear operators between them. Here we can alternatively take the objects to be the tuple $(\mathcal{H}, \langle -, - \rangle)$.
3. The category Set admits sets as objects and functions between sets as the morphisms.
4. The product category $C \times \mathcal{D}$ holds as objects tuples $(a \in C, b \in \mathcal{D})$ and as morphisms tuples of maps $(f \in \text{Hom}_C(a_1, a_2), g \in \text{Hom}_{\mathcal{D}}(b_1, b_2))$.

We can take categories and associate them with each other by use of a *functor*.

Definition 1.1.3. A *functor* F between the categories C and $\mathcal{D}$, denoted as $F : C \rightarrow \mathcal{D}$ assigns to each object $A \in C$ an object $F(A) \in \mathcal{D}$ as well as to every morphism in $f : A \rightarrow B \in \text{Hom}_C$ a morphism $F(f) : F(A) \rightarrow F(B) \in \text{Hom}_{\mathcal{D}}$. Additionally, this must satisfy

$$F(\text{id}_A) = \text{id}_{F(A)}$$

as well as one of the cases

$$F(g \circ f) = F(g) \circ F(f) \quad \text{in the } \textit{covariant} \text{ case} \quad (1.1)$$

or

$$F(g \circ f) = F(f) \circ F(g) \quad \text{in the } \textit{contravariant} \text{ case.} \quad (1.2)$$

We can also relate functor with each other through maps between them. An important relation here will be *naturality*:

Definition 1.1.4. A transformation τ between functors $F, G : C \rightarrow \mathcal{D}$ is *natural* if the diagram commutes for all $A \in \text{ob}(C)$ and $f \in \text{Hom}_C(A, B)$:

$$\begin{array}{ccc} F(A) & \xrightarrow{\tau_A} & G(A) \\ \downarrow F(f) & & \downarrow G(f) \\ F(B) & \xrightarrow{\tau_B} & G(B) \end{array}$$

Example 1.1.5. As an immediate application, consider the category of Hilbert spaces fdHilb and assign it a *forgetfull* functor $\omega : \text{fdHilb} \rightarrow \text{fdVect}_k$ that takes each Hilbert space to its underlying vector space, and each bounded linear operator to its linear operator between vector spaces.

We will now explore how to use category theory to model quantum processes. We want this to be able to do a list of things. First, given two systems A and B , we want the combined system to be an object as well. Given processes, i.e. operators on each individual system $f : A \rightarrow A', g : B \rightarrow B'$, there should be a combined process. Additionally, structures such as entanglement and no-cloning should be readily apparent in this theory.

Consider the tensor product $\otimes_C$ of vector spaces V, W . We can define this using a *universal property*. This should motivate the following definition:

Do we need to give an intro to these properties?

Definition 1.1.6. A *monoidal category* C is a category that admits a unit object $I \in ob(C)$ and a bifunctor $\otimes : C \times C \rightarrow C$, as well as 3 constraining isomorphisms

$$\rho_A : A \otimes I \cong A \quad \lambda_B : I \otimes B \cong B \quad \alpha_{ABC} : (A \otimes B) \otimes C \cong A \otimes (B \otimes C)$$

such that the following diagrams commute:

$$\begin{array}{ccc} (A \otimes I) \otimes B & \xrightarrow{\alpha_{A,I,B}} & A \otimes (I \otimes B) \\ \rho_A \otimes id_B \searrow & & \swarrow id_A \otimes \lambda_B \\ & A \otimes B & \end{array} \quad \begin{array}{ccc} ((A \otimes B) \otimes C) \otimes D & \xrightarrow{\alpha_{A,B,C} id_D} & (A \otimes (B \otimes C)) \otimes D \\ \alpha_{A \otimes B, C, D} \downarrow & & \downarrow \alpha_{A, B \otimes C, D} \\ (A \otimes B) \otimes (C \otimes D) & & A \otimes ((B \otimes C) \otimes D) \\ \alpha_{A, B, C \otimes D} \searrow & & \swarrow id_A \otimes \alpha_{B, C, D} \\ & A \otimes (B \otimes (C \otimes D)) & \end{array}$$

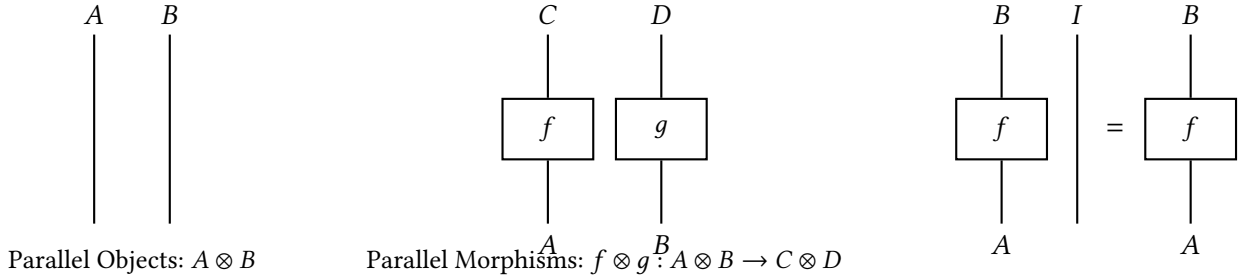
For simplicity we denote a monoidal category as the sextuple $(C, \otimes, I, \alpha, \lambda, \rho)$.

Note that because we defined $- \otimes - : C \times C \rightarrow C$ to be a *(bi)functor*, we immediately gain the following relations by functoriality:

$$(f \otimes g) \circ (f' \otimes g') = (f \circ f') \otimes (g \circ g')$$

$$id_{A \otimes B} = id_A \otimes id_B$$

Graphical Notation 1.1.7. We denote the elements of this Category by simple lines, and the morphisms as boxes between these lines. Since the tensor product corresponds to parallel execution, we describe tensored objects as parallel lines and tensored operators as parallel squares.



The commuting condition in definition 1.1.6 is an instance of the *MacLane-coherence-theorem* [maclane?] (for monoidal categories). This essentially asserts that no matter how many objects we look at, their tensoring are isomorphic to each other without specifying another isomorphism on higher spaces, but instead reusing the α associator on 3 strands is enough.

Example 1.1.8. Some examples of this tensor product here:

1. The category $\text{Rep}(G)$ is monoidal and has a tensor product by ??:

$$(V \otimes_k W, \rho_{V \otimes_k W})$$

and on morphisms $f \in \text{Hom}_G(V_1, V_2), g \in \text{Hom}_G(W_1, W_2)$ this gives

$$(f \otimes g)(v \otimes w) = f(v) \otimes g(w) \in \text{Hom}_G(V_1 \otimes_k W_1, V_2 \otimes_k W_2)$$

2. Take for example the Fib anyon theory. We can take products of anyons $a \otimes b$ and we can also identify the vacuum type I over $a \otimes I = a$ and so on.

Definition 1.1.9. Let $(C, \otimes_C, 1_C)$ and $(D, \otimes_D, 1_D)$ be monoidal categories. A *monoidal functor* is a functor $F : C \rightarrow D$

Definition 1.1.10. Let $(\mathcal{V}, \otimes_{\mathcal{V}}, I, \alpha, \lambda, \rho)$ be a monoidal category. A category C *enriched over $\mathcal{V}$* (or a $\mathcal{V}$ -category) consists of

1. for every $A, B \in \text{ob}(C)$, a *hom-object* $\text{Hom}_C(A, B) \in \mathcal{V}$ (as opposed to Set in normal category),
2. for every triple $A, B, C \in \text{ob}(C)$, a *composition morphism* in $\mathcal{V}$

$$\circ_{A,B,C} : \text{Hom}_C(B, C) \otimes_{\mathcal{V}} \text{Hom}_C(A, B) \longrightarrow \text{Hom}_C(A, C),$$

3. for every $A \in \text{ob}(C)$, a *unit morphism* in $\mathcal{V}$

$$j_A : I \longrightarrow \text{Hom}_C(A, A),$$

This immediately gives us another definition:

$$\text{Hom}_C(-, X) : C \rightarrow \mathcal{V} \quad \text{is a contravariant functor.}$$

Definition 1.1.11. A *braided monoidal category* is a monoidal category $(C, \otimes, I, \alpha, \lambda, \rho)$ equipped with an additional isomorphism, the *braiding*

$$\beta_{A,B} : A \otimes B \cong B \otimes A,$$

on A and B satisfying

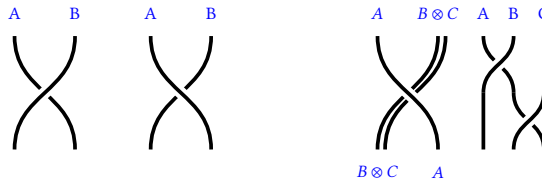
$$\beta_{A,B \otimes C} = (\text{id}_A \otimes \beta_{A,C}) \circ (\beta_{A,B} \otimes \text{id}_C) \quad \text{and} \quad \beta_{A \otimes B, C} = (\beta_{A,C} \otimes \text{id}_B) \circ (\text{id}_A \otimes \beta_{B,C}). \quad (1.3)$$

As a consequence, the *hexagon* relations commute for all $A, B, C \in C$:

$$\begin{array}{ccc} (A \otimes B) \otimes C \xrightarrow{\alpha} A \otimes (B \otimes C) \xrightarrow{\beta_{A,B \otimes C}} (B \otimes C) \otimes A & A \otimes (B \otimes C) \xrightarrow{\alpha^{-1}} (A \otimes B) \otimes C \xrightarrow{\beta} C \otimes (A \otimes B) \\ \beta_{A,B} \otimes \text{id}_C \downarrow & \downarrow \alpha & \downarrow \beta & \downarrow \alpha^{-1} \\ (B \otimes A) \otimes C \xrightarrow{\alpha} B \otimes (A \otimes C) \xrightarrow{\text{id}_B \otimes \beta_{A,C}} B \otimes (C \otimes A) & A \otimes (C \otimes B) \xrightarrow{\alpha^{-1}} (A \otimes C) \otimes B \xrightarrow{\beta} (C \otimes A) \otimes B \end{array}$$

Definition 1.1.12. A braided monoidal category is *symmetric* if $\beta_{B,A} \circ \beta_{A,B} = \text{id}_{A \otimes B}$ for all A, B .

Graphical Notation 1.1.13. Now the graphical calculus changes, as we need to consider whether the braidings defined by β are over or under. We denote the simple braidings as



Check for the correct definition of the ambient space here.

If we want to use contractions between matchings legs, and overall want the ability to take duals just like in FDCVec , we need both a concept of duality and the evaluation maps. This is also the categorification of the *creation* and *annihilation* maps from $??$. Let this motivate the following definition:

Definition 1.1.14. A monoidal category $(C, \otimes, 1)$ is *rigid* if, for every $A \in C$, there exists a *dual object* A^* with the following maps:

$$\text{ev}_A : A^* \otimes A \rightarrow 1 \quad \text{and} \quad \text{coev}_A : 1 \rightarrow A \otimes A^*$$

subject to the diagrammatic constraints

$$\begin{array}{c} A \xrightarrow{\lambda_A} 1 \otimes A \xrightarrow{\text{coev}_A \otimes \text{id}_A} (A \otimes A^*) \otimes A \xrightarrow{\alpha_{A,A^*,A}} A \otimes (A^* \otimes A) \xrightarrow{\text{id}_A \otimes \text{ev}_A} A \otimes 1 \xrightarrow{\rho_A} A \\ \downarrow \text{id}_A \uparrow \\ A^* \xrightarrow{\rho_{A^*}} A^* \otimes 1 \xrightarrow{\text{id}_{A^*} \otimes \text{coev}_A} A^* \otimes (A \otimes A^*) \xrightarrow{\alpha_{A^*,A,A^*}} (A^* \otimes A) \otimes A^* \xrightarrow{\text{ev}_A \otimes \text{id}_{A^*}} 1 \otimes A^* \xrightarrow{\lambda_{A^*}} A^* \\ \downarrow \text{id}_{A^*} \uparrow \end{array}$$

Example 1.1.15. In fdVect_k , the dual object for every vector space V is given by its *dual-space* $V^* := \text{Hom}(V, k)$. The evaluation and coevaluation maps here correspond to

$$\text{ev}_V : V \otimes V^* \rightarrow k,$$

$$\text{coev}_V : k \rightarrow V^* \otimes V,$$

Graphical Notation 1.1.16. Now we have to introduce this notion of duality on the level of the graphical calculus. We denote an object in the category by an oriented arrow, and denote its dual object by the reverse.

Do these need to point up or down?

$$\begin{array}{c} \uparrow A \\ \downarrow A^* \end{array}$$

Similarly the ev_A and coev_A maps induce a graphical meaning:

$$\text{coev}_A : \begin{array}{c} A \\ \text{---} \text{---} \text{---} \end{array} \begin{array}{c} A^* \\ \text{---} \text{---} \end{array} \quad \text{and} \quad \text{ev}_A : \begin{array}{c} A^* \\ \text{---} \text{---} \end{array} \begin{array}{c} A \\ \text{---} \text{---} \end{array}.$$

Observe that in the graphical calculus, we denoted this as a bent arrow from $A \rightarrow A^*$. In fact, we gain the following identities:

$$\text{Hom}_C(X \otimes A, Y) \cong \text{Hom}_C(X, A^* \otimes Y) : f \mapsto$$

Further the natural observation that $A \cong A^{**}$ holds should be encoded in these categories.

Definition 1.1.17. A *pivotal structure* on a rigid monoidal category is given by a natural isomorphism $\text{id} \rightarrow (-)^{**}$, i.e. a collection of isomorphisms that identify $\{A \rightarrow A^{**}\}$.

This allows us to define a notion of a traced category.

Definition 1.1.18. We can define a notion of a *right trace* on any endomorphism $V \rightarrow V$ in such categories:

$$\text{tr}_R = 1 \xrightarrow{\text{coev}_A} A \otimes A^* \xrightarrow{(\delta_A \circ (-) \otimes \text{id}_{A^*})} A^{**} \otimes A^* \xrightarrow{\text{ev}_{A^*}} 1$$

Definition 1.1.19. A pivotal rigid braided monoidal category is called *spherical* if

$$\text{tr}_R = \text{tr}_L$$

and we can further define the *quantum dimension* of an object A as

$$\dim(A) = \text{tr}(\text{id}_A) \tag{1.4}$$

Definition 1.1.20. A *ribbon* category is a braided monoidal rigid category $(C, \otimes, 1, \alpha, \gamma, \rho, \beta, \text{ev}, \text{coev})$ with maps $\theta_A : A \rightarrow A$ that realizes a *self-braiding* of the object. This is subject to

$$\theta_{A \otimes B} = (\theta_A \otimes \theta_B) \circ \beta_{B,A} \circ \beta_{A,B}$$

Definition 1.1.21. Given a linear category C with objects $\{A_j\}_{j=1}^N \subseteq \text{ob}(C)$, an object $\bigoplus_{i=1}^n A_i \in C$ is called the *direct sum* if it is equipped with the injection and projection map

$$\iota_i : A_j \rightarrow \bigoplus A_i \quad \text{and} \quad \pi_i : \bigoplus A_j \rightarrow A_i$$

such that

$$\pi_i \circ \iota_i = \text{id}_{A_i} \quad \forall i \quad \text{and} \quad \sum_i \iota_i \pi_i = \text{id}_{\bigoplus A_i}$$

A great resource on semisimple categories [Bai17].

Definition 1.1.22. A *semisimple* category¹ is a category in which every object $S \in \text{ob}(C)$ decomposes as a finite direct sum

$$S \cong \bigoplus_i S_i, \quad S_i \text{ simple} \quad (1.5)$$

where a *simple* object is an object, where for any other object A that admits an injective map, the object is either the 0 morphism or an isomorphism to S . If we group these simples by isomorphism, we can restate (1.5) equivalently as

$$S \cong \bigoplus_i S_i^{\oplus n_i}$$

where $n_i = \dim \text{Hom}_C(S_i, S)$. Note that by our assumption of fdVect_k -enrichment, all the n_i s are finite. Further denote the set of simple objects by $\text{Irr}(C) \subseteq \text{ob}(C)$.

Example 1.1.23.

In $\text{Rep}(G)$, we know that either the representations are simple (Irreps) or they decompose as some direct sum representation.

In Anyonic theories, we primarily only consider simple objects $a, b, c \dots$ and tensor products between them. Then we assign each combination of anyons a fusion outcome $a \otimes b = c \oplus d \oplus \dots$.

Lemma 1.1.24 (Schur's lemma [Sch05]). *If A and B are simple objects in a linear category over an algebraically closed field k (this is always $\mathbb{C}$ in our case), then it holds that*

$$\text{Hom}(A, B) = \begin{cases} k \cdot \text{id}_S, & \text{if } A \cong B; \\ 0, & \text{if } A \not\cong B. \end{cases}$$

Proof [IR19, Theorem 7.5].

Now assume that we take a semisimple category that is also monoidal. By semisimplicity, any of the objects of the form $R_a \otimes R_b$ are isomorphic to $\bigoplus_c R_c$ where $R_a, R_b, \{R_c\}$ are all simple. A simple consequence of Lemma 1.1.24 produces

$$\dim \text{Hom}_C(R_a \otimes R_b, R_c) = \dim \text{Hom}_C\left(\bigoplus_c N_d^{ab} R_d, R_c\right) = \dim \bigoplus_d \text{Hom}_C(R_d, R_c)^{\oplus N_d^{ab}} \stackrel{\text{lemma 1.1.24}}{=} \dim k^{N_c^{ab}}$$

Call this isomorphism $\Phi : \bigoplus_{c,\mu} R_c \rightarrow R_a \otimes R_b$. Then we can extend projection and injection maps in dependence of the tensor product by the following diagram:

$$\begin{array}{ccccc} \bigoplus_c R_c & \xrightarrow{\Phi} & R_a \otimes R_b & \xrightarrow{\Phi^{-1}} & \bigoplus_c R_c \\ \uparrow \iota_c & \nearrow \exists \iota_{c,\mu}^{ab} & & \searrow \exists \pi_{ab}^{c,\mu} & \downarrow \pi_c \\ R_c & & & & R_c \end{array} \quad (1.6)$$

Furthermore, these admit similar constraints as the direct sum:

$$\pi_{ab}^{c,\mu} \iota_{d,\nu}^{ab} = \pi_{c,\mu} \Phi^{-1} \Phi \iota_{d,\nu} = \delta_{cd} \delta_{\mu\nu} \text{id}_{R_c} \quad (1.7)$$

and

$$\sum_{c,\mu} \iota_{c,\mu}^{ab} \pi_{ab}^{c,\mu} = \Phi \left(\sum_{c,\mu} \iota_{c,\mu} \pi_{c,\mu} \right) \Phi^{-1} = \Phi \text{id}_{\bigoplus_c R_c} \Phi^{-1} = \text{id}_{R_a \otimes R_b} \quad (1.8)$$

We take these maps as above and claim they form a basis of the spaces $\text{Hom}_C(a \otimes b, c)$ and $\text{Hom}_C(c, a \otimes b)$.

¹There are multiple definitions of semisimplicity and semisimple categories, and here we are using the definition of *object simplicity*. A review of these definitions is given in [Bai17].

this is vector space intuition, not sure if this works. Maybe we need to introduce monomorphism? Need to review this definition for general categories
Enrichment first.
Is this just the first part of schurs lemma?

Proof. First we show linear independence. Let $\{x_\mu\}_1^{N_c^{ab}}$ be a set of scalars and suppose that $\sum_\mu x_\mu \iota_{c,\mu}^{ab} = 0$. Observe that

$$\pi_{c,\mu}^{ab} \left(\sum_\mu x_\mu \iota_{c,\mu}^{ab} \right) = \sum_\mu x_\mu \delta_{\mu,\nu} \text{id}_{R_c} = x_\nu \text{id}_{R_c}$$

and since $\text{id}_{R_c} \neq 0$ these form a linearly independent set.

Now let $f \in \text{Hom}_C(R_c, R_a \otimes R_b)$ be an arbitrary morphism. By (1.8), we can write

$$f = \text{id}_{R_a \otimes R_b} \circ f = \sum_{d,v} \iota_{d,v}^{ab} (\pi_{ab}^{d,v} \circ f \in \text{Hom}_C(R_c, R_d)) \stackrel{\text{Lemma 1.1.24}}{=} \sum_{d,v} \iota_{d,v}^{ab} \delta_{R_c \cong R_d} \lambda_v \text{id}_{R_c} = \sum_v \lambda_v \iota_{c,v}^{ab}$$

and f is an arbitrary linear combination and thus

$$\text{span}_{\mathbb{C}} \left\{ \iota_{d,v}^{ab} \right\}_{v=1}^{N_c^{ab}} \cong \text{Hom}_C(R_c, R_a \otimes R_b)$$

and the statement follows dually for $\text{Hom}_C(R_a \otimes R_b, R_c)$. $\square$

If we look at Hom spaces of a higher order such as $\text{Hom}_C(R_d, (R_a \otimes R_b) \otimes R_c)$, we have to choose a bracketing of the tensor products since we can only associate $(R_a \otimes R_b) \otimes R_c \cong R_a \otimes (R_b \otimes R_c)$. This induces a pullback ²

$$\alpha^* : \text{Hom}_C(R_d, (R_a \otimes R_b \otimes R_c)) \cong \text{Hom}_C(R_d, R_a \otimes (R_b \otimes R_c)).$$

We can also define a basis for these extended spaces. In fact, let's generalize this to the space $\text{Hom}_C(R_b, \bigotimes_{i=1}^n R_{a_i})$. Given an ONB ι for the space $\text{Hom}_C(R_b, \bigotimes_{i=1}^{n-1} R_{a_i})$ we can extend it to a basis on the larger space by taking

$$\iota^n = (\iota^{n-1} \otimes \text{id}_{R_{a_n}}) \circ \iota_{b,\mu}^{c,a_n}$$

Definition 1.1.25. A k -enriched category C is called semisimple if every object X is isomorphic to a finite sum of simple objects. Let $\{R_a\}_a$ be representatives of the isomorphism classes of simple categories (and assume algebraic closure / absolute simplicity). Then every X admits as *isotypic* decomposition as

$$X \cong \bigoplus_{a \in L} \text{Hom}_C(R_a, X) \otimes_k R_a$$

Now we would like the category that has all of these statements combined:

Definition 1.1.26. A *fusion-category* is a semisimple, fdVect_k -enriched rigid braided monoidal category, where each simple object has only *finitely many* isomorphism classes.

Remark 1.1.27. While each representation category $\text{Rep}(G)$ forms a fusion category for finite G , this is not the case for infinite G such as $\text{SU}(2)$ or $\text{U}(1)$. These violate the condition that there are finitely many simples. We can remedy this by instead considering a deformation of the underlying Lie-algebra as $U_q(\mathfrak{sl}_2)$.

1.1.1 Categorical Quantum Mechanics

By a result of Abramsky [Abr12], building on the theory of coecke et al., we can state the no-cloning theorem in categorical form: This asserts that there is no diagonal $\delta : V \rightarrow V \otimes V$ such that the following diagram commutes:

$$\begin{array}{ccc} V & \xrightarrow{f} & W \\ \delta \downarrow & & \downarrow \delta \\ V \otimes V & \xrightarrow{f \otimes f} & W \otimes W \end{array}$$

²We will later see this (or where i put it) that this pullback can be assigned a matrix, and through semisimplicity sort of spans the entire space. This is an instance of a lifting through the yoneda lemma, i.e. we can define what α is doing by evaluating the action on finitely many points, and extending this through semisimplicity to the full space.

1.1.2 Tensor & Fusion Categories

1.2 DECOMPOSITIONS FOR CATEGORICAL NETWORKS

This chapter generalizes the decomposition in [SV12, Equation 27] to both Fusion Categories and the Representation categories of compact groups. Note that in [DH25], such a generalization is given for representations of finite groups, which does not include $SU(2)$. However I do not yet know if we should do so, since I not yet sure how to argue the F moves here. If we instead take the deformation $U_q(sl_2)$ we get finiteness (and fusion cats?), unfortunately $U_q(sl_2)$ is not a group.

Lemma 1.2.1 (Isotypic Decomposition). *Let C be a k -enriched semisimple category and $V \in \text{fdVect}_k$. We now give the concept of a **degeneracy space**, where we give a ??? notion of the n -fold direct sum:*

$$V \otimes^k R_a \cong R_a^{\oplus \dim V}$$

How do we make this canonical? by choosing $V = \text{Hom}_C(R_a, V)$.

Lemma 1.2.2. *For a vector space W , there is an isomorphism*

$$\bigoplus^m W \simeq \mathbb{C}^m \otimes W.$$

Proof. Let $e_{i \in [m]}$ be the standard basis on $\mathbb{C}^m$. Let

$$\Phi : \mathbb{C}^m \otimes W \rightarrow \bigoplus^m W$$

be defined by

$$\phi(e_i \otimes w) = (0, \dots, w, \dots, 0)$$

and linear, thus

$$\Phi \left(\sum_i c_i e_i \otimes w_i \right) = (c_1 w_1, \dots, c_m w_m).$$

is clearly bijective, and we have our desired isomorphism. $\square$

Lemma 1.2.3. *Let D be a finite dimensional k -vector space and let V be an object of a k -linear category C . Then*

$$D \otimes V \cong V^{\oplus \dim D}$$

Proof. Let $n = \dim D$. Take a basis $\{e_i\}$ for D . This yields an isomorphism

$$D \cong \bigoplus_{i=1}^n k \cdot e_i \cong k^n$$

Since in a k -linear category the tensor product is additive (should we prove this?), we can distribute

$$D \otimes W \cong \bigoplus_1^n (k \cdot e_i \otimes W) \cong \bigoplus W = W^{\oplus n}$$

$\square$

Lemma 1.2.4 (Isotypic Decomposition). *Let C be a semisimple k -linear category with simple objects $\{R_a\}$. Then every object $V \in C$ admits a canonical decomposition*

$$V \cong \bigoplus_a D_a \otimes R_a. \tag{1.9}$$

where we refer to $D_a := \text{Hom}_C(R_a, V)$ as the degeneracy of the space.

Proof. Since C is semisimple, there exist integers $m_a \geq 0$ such that

$$V \cong \bigoplus_{a \in L} \bigoplus_{i=1}^{D_a^V} a \cong \bigoplus_{a \in L} R_a^{\oplus m_a}.$$

Applying the additive functor $\text{Hom}_C(R_b, -)$ yields

$$\text{Hom}_C(R_b, V) \cong \bigoplus_{a \in L} \text{Hom}_C(R_b, R_a^{\oplus m_a}) \cong \bigoplus_{a \in L} \text{Hom}_C(R_b, R_a)^{\oplus m_a}.$$

Then, by Schurs lemma

$$\bigoplus_{a \in L} \text{Hom}_C(R_b, R_a)^{\oplus m_a} = k^{\oplus m_b}$$

□

So we actually need this: Maybe this is simpler if we always let the degeneracy be C ?

Lemma 1.2.5. *Let C be a k -linear category and let D, E be finite-dimensional k -vector spaces. For any objects $R, S \in C$, there is a natural isomorphism*

$$\text{Hom}_C(D \otimes R, E \otimes S) \cong \text{Hom}_k(D, E) \otimes \text{Hom}_C(R, S).$$

Proof. Let $m = \dim_k D$ and $n = \dim_k E$. Proof by isomorphism chains:

$$\begin{aligned} \text{Hom}_C(D \otimes R, E \otimes S) &\cong \text{Hom}_C(R^{\oplus m}, S^{\oplus n}) && \text{by lemma 1.2.2} \\ &\cong \bigoplus_{i=1}^m \bigoplus_{j=1}^n \text{Hom}_C(R, S) && \text{by additivity of } \text{Hom}_C \\ &\cong k^{mn} \otimes \text{Hom}_C(R, S) && \text{by lemma 1.2.2} \\ &\cong \text{Hom}_k(D, E) \otimes \text{Hom}_C(R, S) && \text{since } \text{Hom}_k(D, E) \cong k^{mn} \end{aligned}$$

(The last as s)

□

Definition 1.2.6 (Coproduct). For a fdVect_k -enriched (and tensored?) category C , and $E \in \text{fdVect}_k$, $X \in C$ the power X^E is the object of C representing the natural isomorphism

$$\text{Hom}_C(Y, X^E) \cong \text{Hom}_k(E, \text{Hom}_C(Y, X))$$

Theorem 1.2.7 (Decomposition of Semisimple Categories). *Let C be a semisimple, monoidal, k -linear category whose tensor product is k -linear in each variable, with simple objects $\{R_a\}_{a \in L}$ and let V_i, W_i be a collection of (not necessarily simple) objects in C . Every morphism*

$$T \in \text{Hom}_C \left(\bigotimes_{i=1}^n V_i, \bigotimes_{i=1}^m W_i \right)$$

admit a decomposition as

$$T \cong \bigoplus_{\hat{a}, \hat{b}} \text{Hom}_k(D_{\hat{a}}, E_{\hat{b}}) \otimes_k \text{Hom}_C(R_{\hat{a}}, R_{\hat{b}}),$$

where $\hat{a} = (a_1, \dots, a_n)$ and P is a k -vector space of dimension ???

Proof.

$$\begin{aligned}
& \text{Hom}_C \left(\bigotimes_i V_i, \bigotimes_j W_j \right) \\
& \cong \text{Hom}_C \left(\bigotimes_i \left(\bigoplus_{a_i} D_{i,a_i} \otimes R_{a_i} \right), \bigotimes_j \left(\bigoplus_{b_j} E_{j,b_j} \otimes R_{b_j} \right) \right) \quad (\text{Lemma 1.2.4}) \\
& \cong \text{Hom}_C \left(\bigoplus_{\hat{a}} D_{\hat{a}} \otimes R_{\hat{a}}, \bigoplus_{\hat{b}} E_{\hat{b}} \otimes R_{\hat{b}} \right) \quad (\text{linearity of } \otimes) \\
& \cong \bigoplus_{\hat{a}, \hat{b}} \text{Hom}_C(D_{\hat{a}} \otimes R_{\hat{a}}, E_{\hat{b}} \otimes R_{\hat{b}}) \quad (\text{additivity of } \text{Hom}_C) \\
& \cong \bigoplus_{\hat{a}, \hat{b}} \text{Hom}(D_{\hat{a}}, E_{\hat{b}}) \otimes \text{Hom}_C(R_{\hat{a}}, R_{\hat{b}}) \quad (\text{Lemma 1.2.5})
\end{aligned}$$

In a basis expansion this then looks like

$$\cong \bigoplus_{\hat{a}, \hat{b}} \sum_{e, f, \mu, \nu} \text{Hom}(D_{\hat{a}}, E_{\hat{b}}) \otimes_k \widehat{T}_{e, f; \mu, \nu}^{\hat{a}, \hat{b}}$$

□

Corollary 1.2.8 (Operators are block matrices [KZ22]). *thm header. An operator*

$$\text{Hom}(V, W) \cong \text{Hom}_C \left(\bigoplus_a R_a^{\oplus d_a}, \bigoplus_b R_b^{\oplus e_b} \right) \cong \bigoplus_{a, b} \text{Hom}(D_a, E_b) \otimes \text{Hom}_C(R_a, R_b)$$

and by Schurs lemma this is just

$$\bigoplus_{a \in \text{Irr}(V) \cap \text{Irr}(W)} \text{Hom}(D_a, E_a) \cong \bigoplus_{a \in \text{Irr}(V) \cap \text{Irr}(W)} \text{Mat}_{d_a \times e_a}(k)$$

Corollary 1.2.9. Let $Q_{\mu, \eta}$ be a basis of $\text{Hom}_C(\bigotimes_{\hat{a}} R_{\hat{a}}, \bigotimes_{\hat{b}} R_{\hat{b}})$ where μ denotes the multiplicities and η the internal fusions. In a multiplicity-free Category we will omit μ as a label.

Example 1.2.10. However, now assign $V_{\frac{1}{2}} \otimes V_{\frac{1}{2}}$ to each leg in the example above. Each leg then becomes $V_0 \oplus V_1 \cong \mathbb{C} \otimes V_0 \oplus \mathbb{C} \otimes V_1$. Then we can start with the decomposition.

$$\begin{aligned}
& \text{Hom}_C \left(\bigoplus_{a, b, c \in (0,1)} (C \otimes C \otimes C) \otimes V_a \otimes V_b \otimes V_c, \bigoplus_{b \in (0,1)} C \otimes V_b \right) \\
& = \bigoplus_{a, b, c, d \in (0,1)} \text{Hom}_C((C \otimes C \otimes C) \otimes (V_a \otimes V_b \otimes V_c), C \otimes V_d) \\
& = \bigoplus_{a, b, c, d} \text{Hom}(C^3, C) \otimes \text{Hom}_{\text{Rep}(SU(2))}(V_a \otimes V_b \otimes V_c, V_d)
\end{aligned}$$

Example 1.2.11. Easier. However, now assign $V_{\frac{1}{2}} \otimes V_{\frac{1}{2}}$ to each leg in the example above. Each leg then becomes $V_0 \oplus V_1 \cong \mathbb{C} \otimes V_0 \oplus \mathbb{C} \otimes V_1$. Then we can start with the decomposition.

$$\begin{aligned}
& \text{Hom}_C \left(\bigoplus_{a, b \in (0,1)} (C \otimes C) \otimes V_a \otimes V_b, \bigoplus_{b \in (0,1)} C \otimes V_b \right) \\
& = \bigoplus_{a, b, c \in (0,1)} \text{Hom}_C((C \otimes C) \otimes (V_a \otimes V_b), C \otimes V_c) \\
& = \bigoplus_{a, b, c} \text{Hom}(C^2, C) \otimes \text{Hom}_{\text{Rep}(SU(2))}(V_a \otimes V_b, V_c)
\end{aligned}$$

Example 1.2.12. Lets look at a tensor with in legs $(\tau \otimes \tau)$ and τ and out legs τ .

$$\begin{aligned}
& \text{Hom}_C(((k \otimes 1) \oplus (k \otimes \tau)) \otimes (k \otimes \tau), k \otimes \tau) = \text{Hom}_C() \\
& \text{Hom}_C(((k \otimes 1) \oplus (k \otimes \tau)) \otimes (k \otimes \tau), k \otimes \tau) \\
& \cong \text{Hom}_C((k \otimes 1 \otimes k \otimes \tau) \oplus (k \otimes \tau \otimes k \otimes \tau), k \otimes \tau) \\
& \cong \text{Hom}_C(k \otimes 1 \otimes k \otimes \tau, k \otimes \tau) \oplus \text{Hom}_C(k \otimes \tau \otimes (k \otimes \tau), k \otimes \tau) \\
& \cong \text{Hom}(k \otimes k, k) \otimes \text{Hom}_C(1 \otimes \tau, \tau) \oplus \text{Hom}(k \otimes k, k) \otimes \text{Hom}_C(\tau \otimes \tau, \tau).
\end{aligned}$$

Example 1.2.13 (Anyonic matrix operator). Finally, consider a $4 \rightarrow 4$ anyon operator with one leg on each side, representing the space $\text{Hom}((\tau \otimes \tau \otimes \tau \otimes \tau), (\tau \otimes \tau \otimes \tau \otimes \tau))$. The full decomposition is $2 + 3\tau$. Hence we obtain

$$\begin{aligned}
& \text{Hom}_C((k^2 \otimes 1) \oplus (k^3 \otimes \tau), (k^2 \otimes 1) \oplus (k^3 \otimes \tau)) . \\
& = \bigoplus_{a,b \in (1,\tau)} \text{Hom}_C(k^{m_a} \otimes a, k^{m_b} \otimes b) = \bigoplus_{a,b \in (1,\tau)} \text{Hom}(k^{m_a}, k^{m_b}) \otimes \text{Hom}_C(a, b)
\end{aligned}$$

where

$$m_1 = 2, \quad m_\tau = 3.$$

For this case, Schurs lemma applies. So we have

$$= \text{Hom}(k^2, k^2) \oplus \text{Hom}(k^3, k^3)$$

which is just a block matrix with indexing $(1_1, 1_2, \tau_1, \tau_2, \tau_3)$.

Example 1.2.14 ($U(1)$ Symmetry). Technically we would want to show here that this is simpler in general. TOOD. Make a theorem? Or a corollary?

$$\begin{aligned}
\mathcal{V}_X \boxtimes \mathcal{V}'_{X'} & \simeq \left(\coprod_{x \in X} \mathcal{V}_{\{x\}} \right) \boxtimes \left(\coprod_{x' \in X'} \mathcal{V}'_{\{x'\}} \right) \quad \text{by (??)} \\
& \simeq \coprod_{x \in X} \left(\mathcal{V}_{\{x\}} \boxtimes \left(\coprod_{x' \in X'} \mathcal{V}'_{\{x'\}} \right) \right) \quad \text{by assumption (i) in first variable} \\
& \simeq \coprod_{x \in X} \coprod_{x' \in X'} \left(\mathcal{V}_{\{x\}} \boxtimes \mathcal{V}'_{\{x'\}} \right) \quad \text{by assumption (i) in second variable} \quad (1.10) \\
& \simeq \coprod_{x \in X} \coprod_{x' \in X'} \left(\iota(\mathcal{V}_x \otimes \mathcal{V}'_{x'}) \right) \quad \text{by assumption (ii)} \\
& \simeq \coprod_{(x,x') \in X \times X'} \left(\iota(\mathcal{V}_x \otimes \mathcal{V}'_{x'}) \right) \quad \text{just to make the base space manifest}
\end{aligned}$$

1.2.1 The Deligne Tensor Product & Product Theories

1.2.2 Algorithms for contracting dual networks

1.3 TENSOR AND FUSION CATEGORIES

Let C and D be linear abelian categories. Then the *Deligne Tensor Product* $C \boxtimes D$ is a linear abelian category with a bifunctor

$$\boxtimes : C \times D \rightarrow C \boxtimes D$$

with morphisms given by

$$\text{Hom}_{C \boxtimes D}(X \boxtimes Y, W \boxtimes V) = \text{Hom}_C(X, W) \otimes \text{Hom}_D(Y, V)$$

Definition 1.3.1 (The Deligne Tensor Product of Categories). Let k be a field, and let $\mathcal{C}$ and $\mathcal{D}$ be two k -linear, finite abelian categories. The **Deligne tensor product**, denoted by $\mathcal{C} \boxtimes \mathcal{D}$, is a k -linear finite abelian category equipped with a k -bilinear functor

$$\boxtimes : \mathcal{C} \times \mathcal{D} \longrightarrow \mathcal{C} \boxtimes \mathcal{D}, \quad (X, Y) \mapsto X \boxtimes Y$$

TODOTOD!!

Definition 1.3.2 (The Category $\text{Rep}_k(G)$). Let G be a group and k a field. The *category of linear representations of G over k* consists of the following data:

- **Objects:** The objects $\text{Rep}_k(G)$ are pairs (V, ρ_V) , where V is a k -vector space and $\rho_V : G \rightarrow \text{GL}(V)$.
- **Morphisms:** For any two objects (V, ρ_V) and (W, ρ_W) in $\text{Rep}_k(G)$, a morphism from V to W is a k -linear transformation $f : V \rightarrow W$ that is G -invariant (also called an *intertwiner*). That is, for every element $g \in G$, the following diagram commutes:

$$\begin{array}{ccc} V & \xrightarrow{\rho_V(g)} & V \\ f \downarrow & & \downarrow f \\ W & \xrightarrow{\rho_W(g)} & W \end{array}$$

In algebraic terms, f must satisfy the condition:

$$f \circ \rho_V(g) = \rho_W(g) \circ f \quad \forall g \in G$$

The set of all such morphisms is denoted by $\text{Hom}_G(V, W)$.

We can further endow $\text{Rep}_k(G)$ with a monoidal structure: For any two objects $(V, \rho_V), (W, \rho_W)$, their tensor product is in $\text{Rep}_k(G)$ by the diagonal action

$$\rho_{(V \otimes W)}(g)(v \otimes w) = \rho_V(g)v \otimes \rho_W(g)w$$

Theorem 1.3.3 (Fusion Representation Category of finite groups). *Let G be a finite group. Then $\text{Rep}(G)$ is a fusion category.*

Theorem 1.3.4. *Let G_1 and G_2 be finite groups and let $\text{Rep}(G_1), \text{Rep}(G_2)$ be their respective symmetry categories. Then there is an equivalence of categories*

$$\text{Rep}(G_1 \times G_2) \simeq \text{Rep}(G_1) \boxtimes \text{Rep}(G_2)$$

Lets give a coordinatized approach to fusion categories:

Definition 1.3.5 (Fusion Category in Vector Form). A *fusion category* is a finite set of labels (sectors?) L and vector spaces V_c^{ab} for all $a, b, c \in L$, as well as the isomorphisms

$$F_d^{abc} : \bigoplus_e V_e^{ab} \otimes V_d^{ec} \rightarrow \bigoplus_e V_d^{ae} \otimes V_e^{bc}$$

such that the pentagon holds.

$$(1.11)$$

$$(1.12)$$

$$(1.13)$$